Jack Sleight

Data Scientist and Software Developer

Glasgow, UK jsleight1 Jack Sleight

Summary

I am a data scientist and software developer with over 7 years of experience. In my current role as an R/R Shiny developer at Audit Scotland, I operate as a lead developer creating web-based applications in R and Python for the analysis of financial and performance audit data. In my previous roles at Fios Genomics, I acted as a senior bioinformatician and developer providing statistical analyses for a range of clients in the pharmaceutical and academic sectors. From my experiences in both the private life science and public audit sectors, I have acquired transferable skills in developing automated data pipelines, transforming complex data sets and creating data-centred user applications.

Skills

Programming

- R
- Python
- SQL
- JavaScript
- Bash

Data science

- Tidyverse, Tidymodels
- Quarto, RMarkdown
- Shiny, Plotly, Bslib, DT
- Numpy, Pandas, Matplotlib, Seaborn

Bioinformatics

- RNA-seq
- Variant calling and haplotype analysis
- Anti-sense oligonucleotide design
- Proteomics
- High-throughput screening development
- Nextflow
- Hypergeometric and gene-set enrichment analysis

Engineering

- Git
- Docker
- CI/CD
- Linux
- Azure: SQL/Blob storage/Container apps/App service/ OpenAI
- R development: Devtools, Testthat, Covr
- Python development: Uv, Poetry, Pytest, Coverage

Experience

Audit Scotland, Glasgow

R/R Shiny Developer

2025-Present

- I design and maintain data applications typically using R, SQL and Python. Projects include:
  - Analysis of general ledger and financial statement data.
  - Automation of Audit Scotland business support functions.
  - Development of custom RAG agent for internal document summarisation.
- I design and maintain data pipelines for extracting, transforming and loading public sector data into structured and semi-structured data sets such as relational databases or JSON file stores, respectively.
- I use Git for version control and developing automated pipelines for maintaining code by adhering to continuous integration (CI) and continuous development (CD) practices.
- I am involved in the maintenance of Azure and Posit infrastructure for developing and hosting applications.
- I lead the development of best practice guidance documentation for R and Python programming.

Fios Genomics, Edinburgh

Senior Bioinformatics Developer **2023-2025**

- I contributed to an internal codebase of over 20 packages typically using R, SQL, Python and JavaScript.
- I was responsible for ensuring reproducibility of statistical analyses using tools such as Docker.
- I scoped development projects to improve and automate bioinformatic analyses.
- I was involved in the communication of new code features via the creation of best practice documentation.

Senior Bioinformatician **2021-2023**

- I acted as a lead bioinformatic data scientist for statistical analysis projects utilising various omics data types. Analyses included:
  - Critical and confounding factors analysis of biological experimental study designs.
  - Analysis of high dimensional data such as gene expression and custom array data sets.
  - Hypergeometric and gene-set enrichment testing of biological pathways.
  - Designing personalised allele-specific ASOs using whole genome sequencing data.
  - Development of machine learning models for predicting drug response using omics data.
- I provided scientific and technical guidance to junior bioinformaticians.
- I was regularly involved in scoping statistical analysis plans for new and existing clients.

Bioinformatician **2019-2021**

- I acted as a junior bioinformatic data scientist performing a range of statistical analyses.

Cancer Research UK Scotland Institute, Glasgow

Laboratory Aide **2015-2016**

- I was responsible for preparation of tissue culture solutions and autoclaving of laboratory waste.

Education

University of Glasgow

MSc - Bioinformatics with Distinction **2018-2019**

- I was awarded a Scottish Funding Council scholarship based on undergraduate academic performance.
- Dissertation: Implementing a pipeline for the detection of copy number variants in hereditary cancer.

University of Strathclyde

BSc - Pharmacology and Biochemistry with First Class Honours **2016-2018**

- I was awarded the Biochemical Society prize for academic achievement.

University of Glasgow

DipHE - Medical Science **2013-2015**

Training and teaching

- 2025 – I attended the Shiny in Production conference run by Jumping Rivers Ltd.
- 2022 – I completed the Exploring Genetic Variation EMBI-EBL virtual course.
- 2021 – I presented a COVID-19 R Shiny application at the Edinburgh Data Visualisation Meetup